CSM
Fall 2026

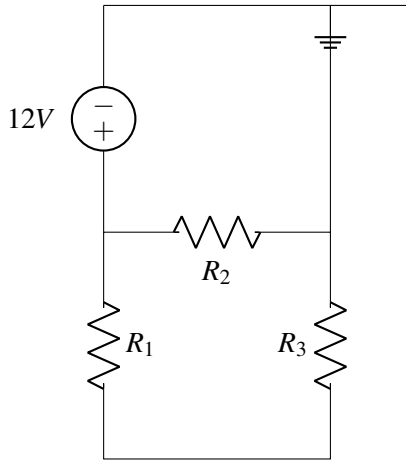
EE 64
Worksheet 2

1. Introduction to KVL, KCL

KVL: Kirchhoff's Voltage Law says that the sum of voltage drops going around a closed network (starting and ending at the same point) is 0

KCL: Kirchhoff's Current Law says that the sum of all currents flowing into a node are equal to the sum of all currents flowing out of a node. This is equivalent to saying that the sum of all currents flowing out of a node (representing flows in opposite direction as negative) is equal to the sum of all currents flowing into a node (again, representing flows in opposite direction as negative) which is equal to zero

Consider the following circuit, with $R_1 = 2\Omega$, $R_2 = 3\Omega$, and $R_3 = 4\Omega$:



We will solve this picture using nothing but KCL and KVL in the following steps.

(a) First, draw a reference point (ground) on the circuit, and label all the nodes V_1, V_2, etc.

(b) Next, draw the +s and -s on the resistors.

(c) Finally, draw arrows indicating the direction of current, labeling them as I_1, I_2, etc.

(d) Now write all the equations given by Ohm's law, KCL and any voltages you can find relative to ground

(e) Now that you have all equations written out, solve them to find all unknown voltages and currents.

(f) Now, we're going to solve the same circuit with Nodal Analysis. Notice that in the previous part, you had many equations are variables (corresponding to both voltage and current). This may make the system more tedious to solve. Instead, Nodal Analysis allows us to write a system of equations that only involves the node voltages.

Start by using the ground and voltage source to find any node voltages which are immediately known.

(g) Then, Ohm's law to write a KCL equation for each node purely in terms of voltages and resistances.

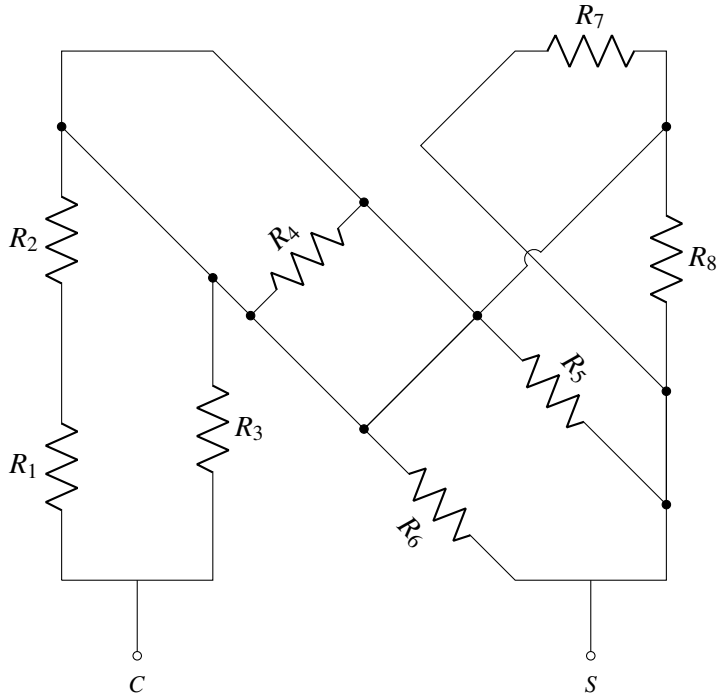
(h) When doing nodal analysis, would it have helped to write a KCL equation at nodes V_1 and V_3? Why or why not?

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2. A Resistor Network Problem

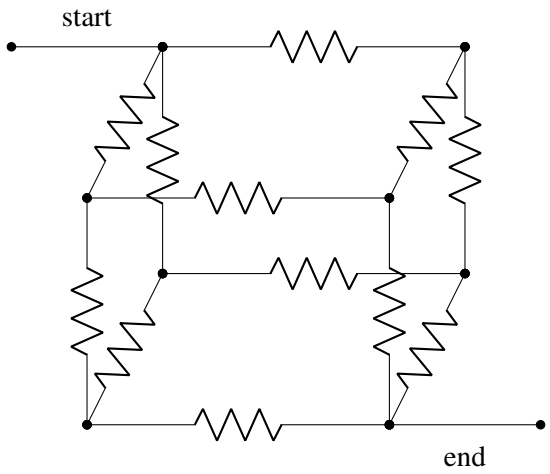
Learning Goal: The goal of this question is to understand equivalent resistance.

(a) Calculate the equivalent resistance between the nodes C and S.



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3. Rubik's Cube (but with resistors!)



In this question, we will guide you through how to apply nodes and resistance equivalence to solve a complex resistor network (like the cubic one above, assuming all resistors are identical with a resistance of R) and find the equivalent resistance between the top back left node and the bottom front right node.

(a) As daunting as this question looks, fortunately we have the power of nodal analysis on our side!

i. First, label all the vertices of this cube with the letters A through H (starting with the top back left vertex as A and the bottom front right vertex as H).

ii. Next, group the vertices from B through H based on the **distances of the shortest paths** along the edges of the cube starting from A. For example, the distance of the shortest path from A to the bottom front left vertex will be 2 (assuming each edge of the cube is 1 unit long).

(b) Now, consider the group of all the vertices v_i that are 1 unit away from A (the starting node in the top back left) and the resistors in and between these pairs of nodes (v_i, A). How are they related to each other (Hint: think in terms of equivalent resistance)?

(c) Now, consider the group of all the vertices v_j that are 2 units away from A. Call it Group II. Let the group in the previous part be Group I. How many edges lead from a vertex in Group I to a vertex in Group II (how many resistors are involved in this case)?

(d) How are Group I and Group II related with each other? Is there anything we can do to simplify the resistor network so far?

(e) Finally, How far away is H (the ending node) from A (the starting node)? How many edges are there that lead from the nodes in Group II to H? Using equivalent resistance, is there anyway we can simplify all the circuits along the edges from the nodes in Group II to H? (Hint: think in terms of symmetry and what you've found in part (b))

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(f) Combining all your findings together, what is the overall resistance between the starting node on the top back left and the ending node on the bottom front right in this cubic resistor network?

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